Quantification: 701_z26c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	10
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2212.2	1656.5	1448.92	px^2
major_axis_length_px	lacuna shape	80.8345	73.4402	31.5243	px
minor_axis_length_px	lacuna shape	43.776	41.6071	14.3109	px
aspect_ratio	lacuna shape	1.9177	1.8485	0.7113	unitless
eccentricity	lacuna shape	0.757	0.8409	0.2081	unitless
solidity	lacuna shape	0.7003	0.6788	0.0841	unitless
roots_count	headline	11.2	10.5	4.0222	count
ring_length_r30_px	headline	450.3	463.5	166.012	px
ring_length_r60_px	reported	1023.1	1017.5	495.929	px
owned_length_px	ownership-dependent	1827.5	1122.29	2152.19	px
edge_count	ownership-dependent	124.4	73	148.802	count
mean_edge_length_px	ownership-dependent	14.6118	14.7273	1.3347	px
perimeter_px	normalised	337.156	314.886	151.985	px
ring_area_r30_px2	normalised	9328.5	9115.5	2703.98	px^2
ring_area_r60_px2	normalised	21741.6	21318	6031.98	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9957	1	0.0135	unitless
ring_density_r30	normalised	0.0480684	0.0506601	0.00925536	px^-1
ring_density_r60	normalised	0.0457018	0.0431809	0.0125177	px^-1
roots_per_100px_perimeter	normalised	3.5545	3.3675	0.8837	per 100 px
ring_attached_length_r30_px	network v2	385.6	372.5	160.113	px
ring_attached_length_r60_px	network v2	731.1	598	500.705	px
ring_length_w_r30_px	network v2	506.051	513.025	185.951	px
ring_length_w_r60_px	network v2	1145.97	1137.22	556.241	px
sholl_crossings_r10	network v2	10.5	11	4.1164	count
sholl_crossings_r20	network v2	11.8	10.5	4.614	count
sholl_crossings_r30	network v2	13.2	12	6.9089	count
loop_count	loops, report only	9.9	1.5	20.3112	count
loop_edge_share	loops, report only	0.1718	0.1338	0.1666	share
ring_width_mean_r30_px	width	6.1446	5.9493	0.762	px

Per field
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measure	value	units
analysed_area_px2	1.02645e+06	px^2
total_skeleton_length_px	46813	px
canalicular_length_density_per_px	0.0456065	px^-1
junction_count	1293	count
junction_density_per_px2	0.00125968	px^-2
lacuna_count	10	count

lacunae_per_px2	9.74e-06	px^-2
skeleton_component_count	818	count
mean_component_length_px	57.2286	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0511931	px^-1
field_density_without_flagged_per_px	0.0438318	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	286	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0456065	px^-1
field_length_density_with_unattached_per_px	0.0458851	px^-1
loop_count	162	count
loop_edge_share	0.1992	share
edge_count	3368	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2156	px
width_median_px	6	px
width_p10_px	2	px
width_p90_px	10	px
width_pixels_used	27069	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1531	85.4004	40.5987	2.1035	0.8798	0.5916	1.0809	183.008	698.054	False
2	2493	118.127	38.0041	3.1083	0.9468	0.6054	-1.3995	230.773	666.071	False
3	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
4	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
5	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
6	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
7	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
8	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
9	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
10	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	11	360	613	403.706	29	13.9209	5.9174
2	12	464	847	1124.78	72	15.6219	5.0033
3	18	570	1337	2205.04	147	15.0003	6.7371
4	10	463	1031	1052	61	17.246	5.5357
5	9	294	549	412.191	32	12.881	6.4769
6	7	307	1004	1119.8	74	15.1324	6.9692
7	7	259	461	303.208	24	12.6337	5.4386
8	8	484	1125	3073.06	216	14.2271	7.4781
9	18	827	2190	7445.73	513	14.5141	5.9087

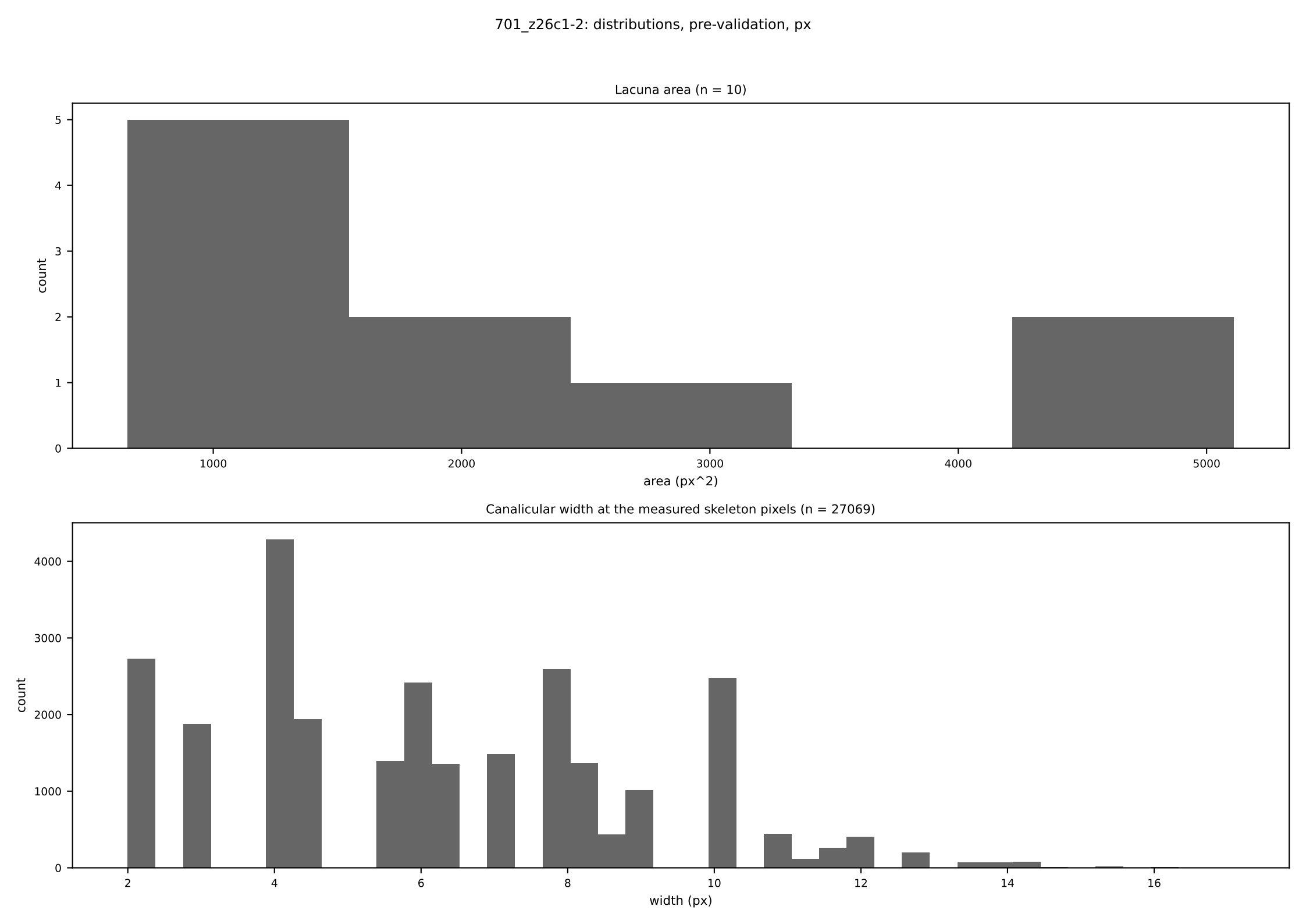
10 12 475 1074 1135.48 76 14.9405 5.9812

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	322.475	7086	14762	1	1	0.0508044	0.0415255	3.4111
2	439.713	9174	16100	1	1	0.0505777	0.0526087	2.7291
3	541.524	14453	32399	1	1	0.0394382	0.0412667	3.324
4	345.031	9057	23202	1	1	0.0511207	0.0444358	2.8983
5	188.244	7262	16365	1	1	0.0404847	0.0335472	4.781
6	273.019	9285	23947	1	1	0.0330641	0.0419259	2.5639
7	133.338	6168	17509	1	1	0.0419909	0.0263293	5.2498
8	215.7	7866	20195	1	1	0.0615306	0.0557069	3.7089
9	605.222	13573	30496	1	1	0.0609298	0.0718127	2.9741
10	307.298	9361	22441	1	0.9574	0.0507424	0.0478588	3.905

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	283	328	409.534	689.428	9	8	9
2	407	569	519.032	946.11	13	12	8
3	536	991	650.541	1510.37	16	18	17
4	338	627	507.019	1156.66	11	9	14
5	218	321	326.936	612.115	6	7	7
6	266	508	346.421	1117.78	6	8	10
7	220	284	295.522	508.889	5	8	4
8	448	1002	540.99	1257.48	11	15	18
9	733	1955	926.411	2447.8	17	20	27
10	407	726	538.103	1213.09	11	13	18


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
0
1
2
3
4
5
1000
2000
3000
4000
5000
Canalicular width at the measured skeleton pixels (n = 27069)
count
width (px)
0
1000
2000
3000
4000
2
4
6
8
10
12
14
16